

Wave 5.2R Cross-Surface Offline Leader Promotion Decision

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Decision

The official `Fw`, `Bw`, and direction-aware `global` TE Curve Verification Pipeline refresh completed on `2026-07-31`. K01 seed `271828` passes the cross-surface offline promotion gate and becomes the Wave 5.2R temporal offline leader for all three surfaces. This is not a TwinCAT runtime or deployable-leader claim.

H08 does not pass global promotion. It remains the preserved non-temporal forward offline specialist because its harmonic and shape behavior is useful, but its `Bw` and `global` raw-error and offset regressions prevent a global leader claim.

The accepted periodic GRU and periodic harmonic MLP are retained unchanged as the non-PINN temporal and non-temporal references. No incumbent artifact was deleted, no accepted registry was overwritten, and the proposed four-global- leader portfolio was not reached because only K01 passed the cross-surface gate.

Evaluation Scope

- Dataset: `polished_dataset`.
- Input mode: setpoints.
- Candidates: `24`, comprising `18` K01/H08 campaign checkpoints and six frozen incumbent controls.
- Curves: `97` forward, `97` backward, and `194` combined global curves.
- Seeds: `314159`, `271828`, and `161803`.
- Policy: multi-index curve-first, with raw error, centered shape, offset, peak-to-peak behavior, derivative fidelity, closure, harmonic amplitude, harmonic phase, seed stability, and visual evidence kept distinct.

Selected-Candidate Evidence

Positive improvement means that the candidate is better than its matched non-PINN lane incumbent.

Lane	Surface	Candidate	Raw MAE [deg]	Offset [deg]	Shape MAE [deg]	Decision
temporal	Fw	K01 <code>271828</code>	0.001358 (+16.07%)	0.000491 (+28.84%)	0.001209 (+12.52%)	pass
temporal	Bw	K01 <code>271828</code>	0.001587 (+13.61%)	0.000564 (+6.93%)	0.001344 (+18.50%)	pass; P2P caveat
temporal	global	K01 <code>271828</code>	0.001464 (+19.12%)	0.000577 (+23.98%)	0.001225 (+19.25%)	pass
non-temporal	Fw	H08 <code>161803</code>	0.001688 (+0.35%)	0.000862 (-4.48%)	0.001320 (+5.04%)	forward specialist

Lane	Surface	Candidate	Raw MAE [deg]	Offset [deg]	Shape MAE [deg]	Decision
non-temporal	Bw	H08 161803	0.001933 (-1.10%)	0.000704 (-15.03%)	0.001658 (+1.13%)	fail global gate
non-temporal	global	H08 314159	0.001871 (-7.90%)	0.000837 (-22.01%)	0.001517 (-2.78%)	fail global gate

K01 also beats the matched GRU on harmonic amplitude, harmonic phase, derivative fidelity, and closure on each selected surface. Its backward peak-to-peak error is 8.100626% , worse than the backward GRU's 6.640831% ; this is retained as an explicit robustness caveat rather than hidden by the aggregate decision.

H08 improves harmonic amplitude and phase substantially against the periodic harmonic MLP, and it preserves useful forward centered-shape behavior. On backward and global surfaces, however, those gains do not compensate for the raw, offset, and associated envelope regressions. The global visual shortlist also shows the larger high-dynamic-condition mismatch.

Seed And Surface Interpretation

K01's three seeds remain ahead of the matched GRU on raw MAE, offset, and centered shape for Fw and global . All three backward K01 seeds also improve raw, offset, and centered shape, while the peak-to-peak caveat remains the main directional weakness. Seed 271828 is selected because it is the best diagnostic-ranked K01 checkpoint on each surface.

H08 is stable in scalar MAE across seeds but not uniformly strong across the full multi-index surface. Its best forward seed is useful, whereas no backward or global seed establishes a balanced win over the matched non-temporal incumbent.

Visual Evidence

- Forward four-model shortlist and curve collages.
 - Backward four-model shortlist and curve collages.
 - Direction-aware global four-model shortlist and curve collages.
 - CVP 1.2 full-resolution curve-payload diagnostics for all 24 candidates.
- The real generated collages were inspected after export. K01 preserves the low-frequency envelope and much of the measured high-frequency structure on both directions. H08 remains plausible on forward curves but exposes larger offset and amplitude mismatch on backward and global high-dynamic examples.

Portfolio State

- The resulting portfolio is deliberately non-destructive
- K01: promoted cross-surface temporal offline leader; TwinCAT runtime pending.
 - H08: retained non-temporal forward offline specialist; not a global leader.
 - periodic GRU: retained accepted non-PINN temporal reference.
 - periodic harmonic MLP: retained accepted non-PINN non-temporal reference.
- The four-global-leader target is therefore not declared achieved. All four families remain available for future research and comparison, but their roles are not falsely flattened into one acceptance class.

Next Work

1. Qualify K01 in the repository-owned TwinCAT replay path before any deployable-leader claim.
2. Keep H08 frozen and diagnose its backward/global offset and envelope defect rather than retraining it blindly.
3. Prepare a separate approval-gated integrated-specialist roadmap covering K01, H08, F01, S01, H04, Stage 10 R00, and Stage 10 S01, with ablations that prove each imported component adds benefit without transferring its known defect.

Artifacts

- Official decision YAML:
`output/analysis/wave_5_2r/offline_leader_cross_surface_track2/official_promotion_decision.yaml` .
- Raw matrix summaries:
`output/validation_checks/track2_reference_comparison/2026-07-31-13-23-15__wave52r_offline_leader_cross_surface_track2_wave52r_offline_leader_cross_surface_promotion_forward/validation_summary.yaml` , `2026-07-31-13-24-17...backward/validation_summary.yaml` , and `2026-07-31-13-25-20...global/validation_summary.yaml` .
- CVP 1.2 diagnostics:
`output/analysis/wave_5_2r/offline_leader_cross_surface_track2/curve_payload_diagnostics/2026-07-31-13-29-13__track2c_curve_payload_diagnostics/` .
- Visual reports: the `forward` , `backward` , and `global` bundles under
`doc/reports/analysis/model_development_waves/wave_5_2/offline_leader_global_promotion/visual_collages/` .